

Growing an Inclusive Community of K-12 CS Education Researchers

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ABSTRACT

A recent study found that there is a litany of unmet needs that are serving as barriers for the CS education research community to grow in depth and breadth, including ensuring that the community is representative of the teachers and students that are studied. Cultivating a diverse, equitable, inclusive, and accessible CSEd research community requires simultaneous bottom-up and top-down alignment on practice standards, professional development, and wellbeing for all constituents that is rooted in politicized trust and collective impact. For this position paper, we engaged in an expository writing process using a confirmatory and elucidating research design to contextualize quantitative and qualitative data reported from our previous study within related work. Our results indicate that there is a variety of researcher-centered, researcher-adjacent, and research-centered barriers in CS education that affect researchers' practice, and personal and professional identities. These results were validated by findings from research in other fields, such as education, psychology, and organizational change. These findings highlight the need for intentional changes to be made, both top-down and bottom-up, to sustain and grow the CS education research community in a way that equitably supports the evolving needs of a diverse set of students as well as the diverse set of researchers who study interventions.

CCS CONCEPTS

• **Social and professional topics** → **Computing education; Computing education programs; Computer science education.**

KEYWORDS

Education research, researchers, systems change, collective impact, capacity, diversity, equity, inclusion, accessibility, position

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1 INTRODUCTION

Issues of representation at all points of the computing pipeline, from early education through professional employment, are well-known to the CSEd research community. In 2014, Pew Research reported that the number of learners in the U.S. identifying as Asian, Black, Hispanic, and other races/ethnicities overtook white learners [34]. As of 2019, the representation of racial/ethnic groups in the U.S. was 47% white, 28% Hispanic, 15% Black and 6% Asian. In California, white K-12 students comprise just 24.7% of all learners [5]. In the U.S. in 2020, only 47% of high schools taught CS, with those schools less likely to have reached learners who: identify as Native American/Alaskan, Black/African American, and/or Hispanic/Latinx; live in rural towns; qualify for free or reduced lunch; are English Language Learners; and/or have disabilities [12]. While residents who identify as women, American Indian/Alaska Native, Black/African-American, Native Hawaiian/Pacific Islander, Multiracial (not Hispanic), or Hispanic have increased their completion of CS bachelor's degrees from 2018 to 2019, during the same period their enrollment in PhD programs decreased 16% and representation among full professors increased just 2% [68, 69]. Of 1,099 publications submitted to three CSEd research venues from 2015–2020, 47% of authors were affiliated with U.S. institutions [36].

CSEd research itself, beyond the makeup of learners and researchers, is underrepresentative of many subjects and methods of study. Considering CSEd research over the last decade, there is still a dearth of K-12 publications [40]. Additionally, one of the challenges of defining empirically-driven best practices in CSEd results from a stark lack of certain forms of research studies (i.e., reproducibility, replication, meta-analysis) due to time pressures, disincentives, and research obstacles [2, 24]. Less than 3% of 2,269 CSEd research studies (2009–2018) were identified as true replication studies, with just six of those focusing on K-12 [26]. Further, only eight of 507 CSEd articles reported the appropriate statistics, study types, and factors to enable meta-analysis [43].

Given the growth in representation of identities historically marginalized in computing among the learner population, the prevalence of U.S. researchers from dominant groups in leadership positions within our CSEd research community, and the lack of certain research subjects and methods, we believe we have to more heavily share the responsibility to address ingrained, systemic "-isms" (e.g., ableism, classism, racism, research elitism, sexism) that may bias the outcomes of our research. The risk of propagating ineffective practices through curriculum, pedagogy, policy, and other interventions leads to eroded trust between practitioners and researchers [4, 28] – trust being a component critical when designing and delivering interventions across lines of difference in power [63].

Diversity, equity, inclusion, and accessibility in CSEd research must be actively negotiated through a critical lens to ensure holistic improvement – for learners, practitioners, study types, and researchers ourselves. In a previous study, we set out to understand the barriers and challenges to conducting equitable and equity-focused CSEd research through a survey of researchers [39]. In this position paper, we summarize those quantitative findings, present additional qualitative findings, and draw from the broader fields of education research and organizational change to assert:

Position statement: *Cultivating a diverse, equitable, inclusive, and accessible CSEd research community requires simultaneous bottom-up and top-down alignment on practice standards, professional development, and wellbeing for all constituents that is rooted in politicized trust and collective impact.*

2 METHODOLOGICAL FRAMING

Within CSEd research (and education research in general), multiple published studies use various approaches to forming and supporting positions. Position papers have been scattered throughout CSEd research to form evidence-based positions for generating thoughtful discussion around problems and challenges requiring attention [29, 65]. Going back as far as 1978, Solntseff made an evidence-based case for using Pascal as the best entry-level language at the time to teach computing [59]. More recently, in 2018 Nelson and Ko used an argumentative presentation in a peer-reviewed ACM ICER paper to claim that theory can inhibit advancement of CSEd research [46]. In addition to the argument presented, the authors provided a set of recommendations to mitigate these inhibitions. Patton describes confirmatory and elucidating research as a method to add "...depth and detail to completed studies that used quantitative data where the statistical results indicate global patterns generalizable across settings or populations" [51, p. 193]. Using this form of research can add meaningful context to quantitative data "...to help make sense out of and interpret survey results."

In a previous study, we performed a systematic literature review on barriers and challenges to CSEd research that is equitable (i.e. across researchers) and equity-focused (i.e. across participant outcomes), which informed the design of a cross-sectional survey that received 72 responses [39]. Survey participants were largely CSEd researchers affiliated with research-oriented (42%) and teaching-oriented (32%) colleges/universities, targeting undergraduate interventions (23%), conducting research within the U.S. (88%), and primarily identifying as white/European descent (73%) and cisgender female (50%) [39].

For this work, we dive deeper into the data collected from the survey using the aforementioned methodology by Patton [51]. In Section 3, we describe condensed results of that study (indicating the percentage of participants who experienced a given barrier) and confirm those results with citations of relevant literature in the closely related fields of education, psychology, and organizational theory. We present the results in categories that help focus the intended target of potential solutions: researcher-centered barriers (e.g., individual and interpersonal), researcher-adjacent barriers (e.g., structural and institutional), and research-centered barriers (e.g., procedural). In Section 4, we adopt an expository writing process to inform the reader about why the barriers elucidated by

our study are important, and how they can be mitigated through existing models for bottom-up and top-down change in education.

3 CONFIRMING AND ELUCIDATING OUR FINDINGS WITH RELEVANT LITERATURE

3.1 Researcher-Centered

3.1.1 Researcher-Centered Barriers. Participants of our previous study [39] indicated a number of individual and interpersonal barriers supported by the literature. A primary factor is time (working more than a 40-hour week, 64%; constraints with teaching vs. research, 53%; increased workload due to peer reviews, 43%; care-taking responsibilities, 40%; personal commitments, 29%) [10, 17]. Participants also described the time-related barriers of balancing administrative and project management tasks, securing and maintaining funding, keeping up with the literature, and conducting rigorous study methodologies (e.g. longitudinal, replication [2, 18, 26]). Participants also indicated stress and strain related to publishing (36%) [10, 32] and tenure or promotion (17%) [16]. The item that received the highest rate of response across the entire survey was "Issues related to COVID-19 pandemic" (n=58, 81%) [19, 45].

Unfortunately, many participants indicated that they experience barriers of low self-efficacy (43%), low sense of belonging (36%), and lack of recognition for research on outreach activities (21%) [9, 18, 31]. Participants also indicated the barriers of a lack of diversity among CSEd researchers (43%) and leaders (40%) [66]. A participant identifying as a Black American shared that they "are not taken as seriously" as the results of their program merit, while another participant shared their desire that "people respected me and my time (as a female) the way they do of male colleagues." Participants shared their experiences of CSEd research as a "clique-y community that doesn't seem to welcome newcomers," of isolation, of a lack of deliberate connection of groups within CSEd research, and of misplaced competition among constituents:

"I would love it if three groups could get along: (a) educational researchers (b) computer scientists (c) instructors/teachers/professors of computing. As far as I can tell, rivalries seem like the softer footfalls than collaborations."

3.1.2 Researcher-Centered Solutions. Researcher-centered solutions for equitable and equity-focused CSEd research include providing development on cultural competence, entwining equity throughout our research, and prioritizing underrepresented topics and audiences while broadening inclusive and precise language. Many of the researcher-centered barriers discussed above are specific to an individual's personal and professional context, within which the CSEd research community can align and advocate for change, but the improvement must ultimately come from the context itself. As described by Brownell and Tanner, "...if change takes time and training, then faculty need to be compensated for their efforts in the form of lower teaching loads, financial benefits, recognition for tenure, teaching awards, or even, at the most basic level, verbal acknowledgment from colleagues and supervisors" [7, p. 340].

Some time-related issues can be solved by scrutinizing our peer review and publication practices, including explicit calls for underrepresented methodologies and areas of focus, clear review criteria,

and reviewer training [26, 53], as demonstrated actively by ACM and ICER [15, 23]. We must also ensure that our practices are inclusive of a broad but rigorous set of writing styles, lived experiences, and research areas [25]. Many of the best practices for cultivating self-efficacy and sense of belonging among learners can also be applied to ourselves: establishing near-peer and senior mentors, normalizing failure (including valuing inconclusive and negative results), defining professional identity, and demonstrating empathy for one another as complex individuals of intersectional identities [60]. Diversity and equity, which leads to belongingness and is related to networking and collaborations (and arguably more research output), will not happen by accident—careful planning and appropriate resources are needed.

3.2 Researcher-Adjacent

3.2.1 Researcher-Adjacent Barriers. Participants noted many structural and institutional barriers to diverse, equitable, inclusive and accessible CSEd research. 36% of participants stated that finding researchers to conduct quality reviews was challenging. This was closely related to barriers to dissemination, including difficulty getting research published (29%), lack of journals for publishing CSEd research (25%), lack of opportunities to contribute to the body of CSEd research knowledge (14%) and lack of publication venues that broadly and consistently value equity-focused research (18%). One participant noted:

"I have been stunned by how arbitrary peer review has become. I increasingly see people who support poorly-constructed research because it supports their personal agenda, and dismiss well-constructed research because it disagrees with their own mental models. The quality and relevance of research published in SIGCSE venues has taken a nosedive. Among other things, the majority of research is particular to large institutions only."

Previous findings surface known issues with peer review in CSEd research [53] that parallel our findings, highlighting the need for people and processes to address this area.

As a new field, our research scholarship output is quickly increasing while the support for dissemination is not keeping pace. Participants also noted a palpable bias towards certain methodologies and research focus areas, including an overall lack of awareness about the many types of CSEd research, which could further prevent important forms of research from being conducted and disseminated. Unsurprisingly, CSEd researchers have indicated beliefs that novel works both carry more inherent value (e.g., likelihood for publishing, citation, funding) and that they more positively influence promotion [2], highlighting a fundamental misunderstanding of the value of replication at best, and/or a bias propagated by the systems that cultivate and advance researchers at worst.

Likewise, one in three participants noted a lack of funding to conduct CSEd research was an issue for them personally as well as the broader community. This impacts conference attendance and publication of open-access articles (which often cost money) [8, 58]. Participants also noted that the funding pipeline is too long, especially when funders require them to envision the research far in advance of the immersion into the study. There is also the problem

of funders requiring a proof of concept prior to getting funding, but researchers need funding to conduct the proof of concept [8]. Notably, participants also found short funding cycles (three years, for example) to be problematic when the project design, development, evaluation, and iteration process is much longer (particularly when scaling is needed).

With respect to a career pathway in CSEd research, participants noted a lack of recognition for (e.g., not counted for tenure or promotion) and acceptance of equity-focused CSEd research (26%). Participants also noted a "lack of institutional recognition of CS education research as research," lack of career pathways outside of academia that value CSEd research graduates' skills and knowledge, and lack of faculty positions in CSEd as supported by Cooper et al.

Traditional, discipline-specific mechanisms for cultivating education researchers do not exist at a robust scale for CSEd, especially at a proportion necessary to complement the rapid growth of the PreK-16+ field. First, while many are emerging, participants indicated a lack in quantity of (35%) and funding for (36%) PhD programs in CSEd. An analysis of 1,099 CSEd publications at three venues from 2015-2020 "considered doctoral consortia (DC) as a measure of where graduate research occurs," [36, p.291] identifying just 13 institutions with greater than two students contributing to the DC and 15 institutions with a single student contributor (among an overall sample of 578 global institutions). Further, CSEd publication growth in the U.S. primarily results from new authors at previously published institutions, indicating a concentration of formal CSEd research training inaccessible to many [17, 36]. This is supported by participants who indicated minimal mentors for CSEd researchers and graduate students (35%), and insufficient professional development (22%) and education research knowledge bases (58%) that are appropriately contextualized to CSEd. The lack of formalized training for CSEd researchers is exacerbated when considering the importance and prevalence of CSEd researchers from outside academia: those in non-profit and for-profit sectors as well as educators and practitioners as investigators.

3.2.2 Researcher-Adjacent Solutions. An aggregation of resources, practice standards and professional development can happen outside of academia and other institutions where CSEd researchers are employed, but many barriers related to wellbeing and career advancement require hyperlocal change. However, the CSEd research community can align bottom-up on proposals for peer review processes, publication venues and criteria, career pathways, and formal training paths for future CSEd researchers.

Organizations can adopt strategies that encourage quality practice standards and a diversity of methodologies from the top down. Gap and meta-analyses provide insight into who and what are being studied, how, and by whom, enabling us to understand trends in the research corpus [26, 37, 62]. These analyses, as well as reproducibility and replication studies, require access to open and transparent datasets [13, 24]. Further, the accessibility and transparency of open data can increase public and practitioner trust [13, 28], as well as enabling researchers to incur citations [54]. The Psychological Science journal implemented Open Science Badges to publicly verify that a publication meets openness standards. Less than two years after the badges were introduced, the number of articles reporting open data increased from 3% to 39% [33]. Repositories that detail intervention

variables and participant characteristics, standardized instruments that demonstrate validity and reliability, and outcome data would accelerate the existing ability of CSEd research to synthesize findings across contexts and respond meaningfully to opportunities, constraints, and needs.

It is imperative that we understand who is participating in CSEd and CSEd research interventions by collecting and reporting demographic data that allow the lived experiences of learners and practitioners to be duly understood, and enable benchmarking across datasets. With demographics (e.g., age, disability status, race/ethnicity, socioeconomic status), we must align on the most inclusive and precise language for a variety of demographic subgroups and how to collect and report it quantitatively while supporting participants to self-describe [50]. To ensure that intervention outcomes are not experienced inequitably by subgroups, demographic data should be reported both disaggregated within a single dimension (e.g., race/ethnicity) and intersectionally across multiple dimensions (e.g., race/ethnicity and gender) [12]. Publication and review processes can adapt to accommodate reporting of this data in overall length requirements, train reviewers on its value and how to assess it, and incentivize and/or require these data be reported.

3.3 Research-Centered

3.3.1 Research-Centered Barriers. Participants noted a total of 28 barriers related to a lack of experience, training, or knowledge in CSEd research for them personally and among the community. One participant noted that they needed "proper training on stats and qualitative analysis" to better conduct research. Others noted a lack of awareness and adoption of practices from other education research fields (58%), a lack of replication, reproducibility, and/or meta-analysis studies (42%), and lack of knowledge of general educational research theory (39%). Participants also noted a lack of breadth across CSEd research topics (32%) and equity-focused CSEd research in the literature (31%). One participant noted a "lack of access to low-cost or free ware or open source ware to facilitate instrument creation and data collection." 36% of respondents stated that there was a lack of equitable representation among identities of participants in studies while 29% reflected that there was a lack of access to diverse participants.

Conferences are a key component for researchers to network, which often forms the basis of future collaborations. Participants (40%) noted that they experience barriers to effective research collaboration, with some noting that they experienced physical barriers to participate in conferences (11%) as well as a lack of equal opportunities for networking (24%). During the time of COVID, this is even more challenging for immunocompromised individuals, who may typically be able to attend conferences but are placed at higher risk for in-person attendance.

The need to validate theories and findings in education through reproduction and replication of studies has gained increasing attention [30, 35, 49]. Ho and Vuong examined economics, political science, psychology and more to determine the need for replication studies [28], referencing a high-visibility study in *Nature* illustrating that 90% of the scientists ($n = 1,576$) noted that there is indeed a "reproducibility crisis" [28, p. 14]. Closer to our community, a 2015 ACM ITiCSE Working Group conducted a meta-analysis of

a set of CSEd papers [30], and their recommendations included promoting validating and replicating studies to garner a more valid understanding of influencing variables.

3.3.2 Research-Centered Solutions. Community leaders and publication venues can encourage researchers to improve the rigor and integrity of their research design by registering research designs and methods in advance of data collection to ensure the hypothesis and analyses remain consistent despite the outcome of the study (i.e., to avoid post-hoc revision, significance chasing, selective reporting, etc.) [4, 24, 27]. One study found that 4-15% of researchers in single-case design studies would drop cases with small effect sizes when submitting for publication. Further, reviewers are likely to favor submissions with larger effect sizes, even if the methodology is the same with different results between studies [57], let alone submissions that publish null or negative results, calling for a realignment of what types of results are valued within research communities. Incentivizing and recognizing open and large-scale collaborative science, participatory design, pre-registration, and reproducibility, replication and meta-analysis studies can encourage CSEd researchers to adopt these methods [7].

Inconsistency in reported statistics within CSEd research, [3, 42, 55, 56] especially when compounded by a lack of standardized instrumentation [37], presents an immediate need and achievable opportunity to improve reporting practices both as individual researchers and as a culture of practice. A minimum viable set of statistics common to other fields [20, 64] can be reported, as appropriate, to aid in synthesizing results and conducting reproducibility, replication, and meta-analysis studies: sample size, mean, standard deviation, confidence interval, scale reliability and validity, power, and effect size. Previous work suggests a number of ways to move towards acceptance of these studies, including incentivizing researchers and starting with smaller steps such as preregistration and training researchers to do more comprehensive reporting [38]. Data that is not found to be significant also holds meaning and should be reported. Further, explaining which analyses are used and why is critical to assess the appropriateness of statistical measures, qualitative protocols, and experimental design [24]. Many such frameworks and repositories exist in other disciplines [20, 47], and within CSEd research already [15, 22, 23, 37, 41]. Moreso than creating more of them, they require awareness building and championing by key figures in the community; ideally, by a unified voice akin to What Works Clearinghouse.

4 DISCUSSION

The solutions we have selected for this paper represent actionable and achievable opportunities that are currently being implemented in other disciplines of education research with a lens customized to the needs and challenges of CSEd researchers. Our position is that many solutions to address known barriers in CSEd research already exist; it is not so much a question of what or how, but who and where. While many barriers require change within the context where CSEd researchers are employed, we can affect change through grassroots alignment as a community that influences the actions taken and resources provided by leaders in the space.

Bottom-up change from all members of the CSEd research community is a fundamental necessity for our continued evolution.

Extending Wenger [67], communities of practice applied within CSEd research can affect transformations internally (e.g. research methodology), externally (e.g. in interventions, curriculum, and pedagogy), and over time (e.g. responsive to new technologies and needs). Fincher and Tenenberg applied Wenger’s dimensions of mutual engagement (dynamic and continuous interactions on issues of shared interest and meaning) in a joint enterprise (emergent projects that members negotiate and hold one another accountable to) through shared repertoire (tools, concepts, etc. that the community has adopted or created) to the lack of scientific paradigm in CSEd research [21]. As presented in Section 3, our scientific paradigm and shared repertoire has evolved substantially; we currently have the opportunity to strengthen our mutual engagement of CSEd research constituents with historically marginalized identities and implement the community-wide accountability central to joint enterprise. Through this perspective of the community role, CSEd research is not a self-contained, academic enterprise, but a dynamic practice centered in the unique lived experiences of researchers and participants alike among a constellation of domains and methodologies applied for real-world change.

Top-down change through clear signals from leaders and trusted entities in CSEd research are a necessary complement to ensure the grassroots efforts are promptly validated, proliferated through the community, and made discoverable for future members. As presented in Section 3, many members of our community experience researcher-centered issues of self-efficacy, belongingness, and professional identity as a CSEd researcher, especially for those who hold intersecting marginalized identities. This compounds with the researcher-adjacent lack of time, training, and structural incentives and support that impede individual researchers’ ability to evolve their practice, even when they desire to, and be rewarded for their accomplishments. Leadership entities hold a unique ability to address the challenges of awareness and pressure that researchers experience by building capacity for, access to, and participation in a culture of improved practice. Capacity-building requires expanding beyond individuals and volunteers within the CSEd research community and ensuring internal processes are funded, led, and implemented in a way that enables systemic change in linking education research and practice [8]. Stakeholders with cross-cutting perspectives and robust resources must play a significant role as central leaders in this systemic change, without enforcing strict hierarchies or lacking accountability to constituents.

Equity-focused, empirical research and interventions require an intimate knowledge of the needs and contexts of participants; so, too, requires the CSEd research community when defining an equitable and sustainable future for ourselves. A successful way to build that knowledge is through participatory design. Vakil et al. argue for the use of participatory design in education research to directly involve members of the community being studied in the design and implementation of the research itself. This is paired with rigorous introspection of researcher positionality to challenge hierarchical power dynamics usually present when researchers enter a community they do not currently belong to, especially across differences in race and socioeconomic status. Politicized trust, a concept where relationships are negotiated along the historical and sociocultural weight that each member’s identities bring and how those identities interact, must be explicitly created and actively maintained,

and must directly challenge assumptions of objectivity [63]. The broader education research community has had time to develop, implement, and iterate on methodologies like design-based research, which emerged in the early 1990s [6, 14], and has since grown into more critical perspectives like politicized trust. Politicized trust is not only beneficial to establish with our participants, but with our peers and leaders in the CSEd research community. Building trust and countering problematic research practices requires that research communities “...must solve the challenge of coordinated action in a decentralized system” [48, p. 2605].

There are a number of models for coordinated action that could be adopted, and in fact have been used in CSEd spaces, to intentionally engage in the process of making our spaces more inclusive. These models include the simultaneous bottom-up and top-down alignment we assert as necessary to drive impactful change. The collective impact (CI) model is defined as “a network of community members, organizations, and institutions who advance equity by learning together, aligning, and integrating their actions to achieve population and systems level change.” [61] Centering equity is key in a CI model, and partners collaboratively engaging in CI each play a distinct role that is supported by a backbone team. Building off of similar strategies employed by collective impact, there are two current examples of simultaneous top-down and bottom-up change in CSEd that can be adapted from interventions for learners to the wellbeing and expertise of the CSEd researchers themselves. Research Practice Partnerships (RPPs) bring together researchers and practitioners to address problems of practice that practitioners experience and that ultimately impact their students [11, 52]. Within K-12 CSEd, there is a recent, rich history of RPPs across the U.S. [44]. Additionally, the U.S. NSF Computer and Information Science and Engineering directorate call for plans to broaden participation in computing (BPC) when submitting proposals for certain research programs [1], with a central review, feedback and measurement process, which has been complimented by BPCNet.org. The website lists a repository and best practices of existing BPC activities contributed by the CSEd community and maintained in partnership with other organizations. Borrowing and learning from these models could provide our community with structured mechanisms for informing, implementing, and measuring change. Future work will explore additional models for collaborative top-down and bottom-up systems change, the degree to which they are relevant to the barriers we know are experienced by CSEd researchers, and the feasibility of implementing the models that are appropriate.

5 COUNTERARGUMENTS AND LIMITATIONS

We imagine possible counterarguments to the above. First, these efforts should emerge organically as the discipline matures, and even that they are already underway. Second, given the rapidly evolving nature of CS and inherent delay as researchers and practitioners translate those developments to learning environments, there are more urgent priorities for the CSEd research community to evaluate novel interventions tied to new technologies, curricula, and pedagogy. Third, engaging in certain solutions outlined requires far greater sacrifice from certain researchers based on their various personal and professional identities and relative privileges. Finally,

certain influences, constraints, challenges, and solutions are not included in this paper.

Each of these is valid and crucial to consider when orienting as a community for collective change. We acknowledge that the proposed requires intentional effort and investment, and evolution on the part of our established institutions. Our CSEd research community has made commendable strides in defining, iterating, and improving a research methodology and corpus; the wealth of references in this paper representing a fraction therein. However, the frequency of issues in the literature and the responses from our survey participants indicates ample opportunity to improve the collective experience and practice. In parallel, the range in relative complexity of solutions outlined, from design and methodology enhancements by individuals to systemic change of entrenched practices that perpetuate inequities, provides a breadth of ways we can take actions that respond to challenges within CSEd research practice and remain agile to the evolving applications of research.

Regarding certain solutions requiring greater sacrifice from some, here, again, the need for systemic change cannot be understated. It is incumbent upon every member of the community to leverage the forms of privilege they have to dismantle and disrupt practices that favor researchers with preparatory privilege, fail to challenge easily accessible yet unrepresentative study participants, and inadequately incentivize and recognize certain topics. These efforts are not separate from or nice-to-have in CSEd research; they are integral to it, and require us to consistently enhance our cultural competence and demonstrate concrete action as accomplices, not just allies, of our colleagues and constituents from historically marginalized groups.

The demographics of our survey respondents do not capture the lived experiences of the intersectional identities of all CSEd researchers, and thus our previous findings do not capture all barriers that researchers face. This snapshot is also indicative of the unrepresentative state of the CSEd research community relative to the diversity of practitioners and learners we aim to support.

6 CONCLUSION

The growing quantity and diversity of learners in PreK-16+ CSEd, and the lack of diversity among CS and CSEd academics and researchers, provide ample justification to adopt bottom-up and top-down alignment on practice standards, professional development, and wellbeing for all constituents in CSEd research that is rooted in politicized trust and collective impact. CSEd research is (comparatively) new enough that researchers, particularly novices and those from historically marginalized groups, are unaware of certain practices and venues, and/or feel unwelcome in certain spaces. While there has been extensive discussion across multiple disciplines regarding research practices to avoid generalizing erroneous or biased results, there is particular urgency in CSEd as the experiences of its learners, practitioners, and researchers continues to be unjust due to systemic exclusion and perpetuation of inequities toward our members from historically marginalized groups. We recognize and have heard enough concerns across the community to warrant a bigger picture view of the challenges that our community faces. If our community wants broader changes to address systemic issues and to increase quality opportunities for and output

from researchers, we need an integrated approach with targeted solutions. If the early solutions we put forth can be more refined with input from the community, we believe these discussions are well worth the time, attention and actions to become a more inclusive community addressing CSEd research for all.

